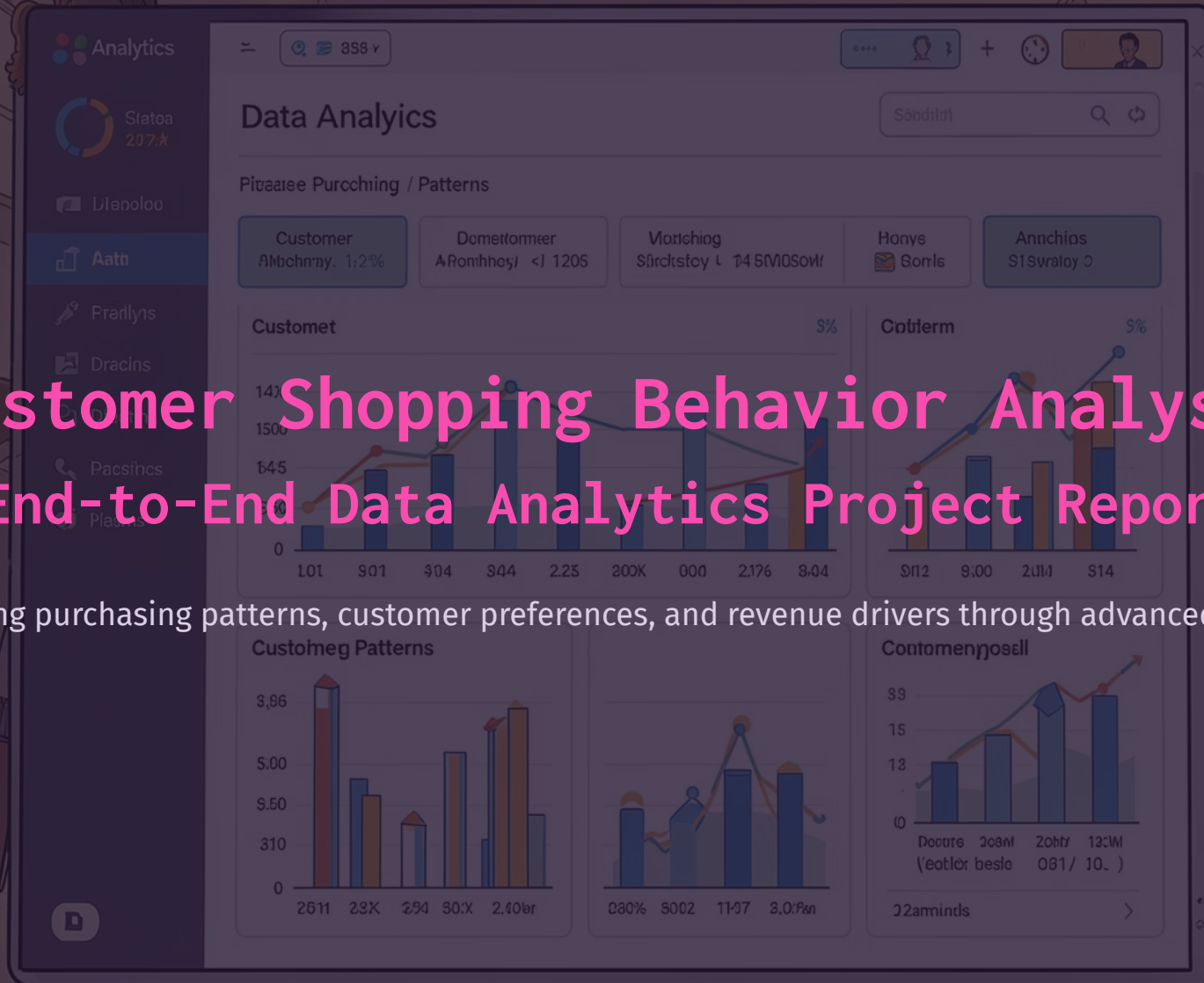


Customer Shopping Behavior Analysis

End-to-End Data Analytics Project Report

Uncovering purchasing patterns, customer preferences, and revenue drivers through advanced analytics.



Executive Summary

Project Overview

Analyzed customer shopping behavior using transactional retail data to uncover purchasing patterns, customer preferences, product performance, and revenue drivers using Python, PostgreSQL, SQL, and Power BI.

Business Problem

Retail businesses generate large amounts of customer transaction data. Without proper analysis, valuable insights regarding customer preferences, spending habits, and product performance remain hidden.

Key Objectives

Understand customer purchasing patterns, Identify high-value customer segments, Evaluate product and category performance, Measure subscription effectiveness, Improve marketing and retention strategies

Dataset Overview & Methodology

Dataset Description

Approximately 3,900 customer transaction records with variables: Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Color, Season, Review Rating, Subscription Status, Shipping Type, Discount Applied, Promo Code Used, Previous Purchases, Payment Method, Frequency of Purchases

Tools & Technologies

- Python: Data cleaning, transformation, feature engineering
- PostgreSQL: Data storage, query execution, BI analysis
- SQL: Data aggregation, customer segmentation, revenue analysis
- Power BI: Dashboard creation, KPI visualization, interactive reporting

Data Preparation & Processing



Data Cleaning

- Checked for missing values
- Verified data consistency
- Standardized column formats
- Removed data irregularities
- Validated numerical fields



Feature Engineering

Created Age Group variable for demographic analysis: 18–25, 26–35, 36–45, 46+ to enable customer segmentation based on age demographics



Database Integration & Validation

Imported cleaned dataset into PostgreSQL using SQLAlchemy, Psycopg2, pgAdmin 4. Used SQL queries to verify row counts, validate data integrity, confirm aggregation results, and ensure consistency between Python and PostgreSQL outputs

Customer Behavior Analysis

1

Revenue Analysis by Gender

Both genders contribute significantly to overall revenue. Spending patterns vary across demographic groups. **Business Value:** Understanding gender-based spending helps businesses design targeted marketing campaigns and product promotions.

2

Customer Segmentation

Segment	Criteria
New Customer	1 Previous Purchase
Returning Customer	2–10 Previous Purchases
Loyal Customer	More than 10 Previous Purchases

Key Finding: Loyal customers contribute a substantial portion of total revenue. Returning customers represent strong growth opportunities.

3

Repeat Purchase Analysis

Customers with more than five previous purchases were evaluated. Repeat buyers generate higher long-term value. Subscription customers demonstrate stronger repeat purchasing behavior.

Product & Category Performance

Top-Selling Categories

Category performance was analyzed using transaction volume and revenue generation.

Key Findings:

1. Clothing
2. Accessories
3. Footwear
4. Outerwear

Clothing consistently generated the highest revenue and sales volume.

Top-Rated Products

Average review ratings were calculated for all products. Products with consistently high ratings provide opportunities for:

- Promotional campaigns
- Inventory prioritization
- Cross-selling strategies

Category Performance Insights

The analysis revealed that product popularity varies significantly across categories.

Business Impact:

- High-performing products should receive greater visibility.
- Inventory allocation should prioritize strong-performing categories.

Subscription & Promotional Analysis

27%

Subscribed

Subscription Status Performance

Customer spending behavior was analyzed based on subscription status. Subscribers show stronger purchasing engagement. **Business Value:** The subscription program contributes positively to customer retention and revenue generation.

73%

Non-Subscribers

Promotional Opportunities

The findings suggest opportunities to: Personalize promotions, Improve campaign targeting, Increase customer lifetime value

Discount Effectiveness Analysis

Discount utilization was evaluated across products. Certain products demonstrate high discount dependency. Price-sensitive customer segments respond positively to promotions. **Business Value:** Strategic discounting can increase conversion rates while maintaining profitability.

Dashboard Insights & KPI Performance

3.9K

Total Customers

\$59.76

Average Purchase Amount

3.75

Average Review Rating
(out of 5)

46+

Top Revenue Demographic
Years

Revenue by Age Group

Customers aged 46+ contribute the highest revenue. **Business Implication:** This demographic represents a highly valuable customer segment and should be prioritized for retention initiatives.

Sales by Age Group

Customers aged 46+ also record the highest purchase frequency. **Business Implication:** Older customers exhibit stronger purchasing behavior and greater loyalty.

Customer Satisfaction

The average review rating of 3.75 indicates moderate customer satisfaction. **Recommendation:** Further investigation should identify products or categories with lower ratings to improve customer experience.

Recommendations & Conclusion

Strategic Recommendations

Customer Retention

- Expand loyalty and rewards programs
- Offer exclusive benefits to repeat and loyal customers

Marketing Optimization

- Personalize campaigns based on age groups and purchase history
- Use customer segmentation for targeted promotions

Subscription Growth

- Introduce subscriber-exclusive discounts
- Promote membership benefits to non-subscribers

Product Strategy

- Increase visibility of top-selling and highly rated products
- Maintain inventory levels for high-demand categories

Revenue Growth

- Focus marketing investments on high-performing demographics
- Leverage purchasing frequency data to recommend products and increase customer lifetime value

Conclusion

This project successfully demonstrates an end-to-end data analytics workflow using Python, PostgreSQL, SQL, and Power BI. Through data preparation, advanced querying, customer segmentation, and interactive dashboard development, valuable insights were generated regarding customer behavior, product performance, subscription engagement, and revenue drivers. The findings provide actionable recommendations that can support data-driven decision-making, improve customer retention, optimize marketing performance, and drive sustainable business growth.